Dendrograms Diagrams to support Hierarchical Clustering

Lab 4

Instructions

This lab assignment exposes you to using Dendrograms diagrams to support Hierarchical Clustering. This process is trying to find groups, where you start with one big group, and you split it apart, split it apart, split it apart until you have individual cases. A dendrogram is a diagram that displays the hierarchical relationship between objects. It is most often created as an output from hierarchical clustering. The essential use of a dendrogram is to work out the best way to allocate objects to clusters. There are three main steps:

1. Access Google Colab by signing into Google
2. Create a new notebook and install and import Python Packages
3. Download dataset and summarize data

Lab Background

For the lab, we will be using the dataset “Cereals.csv” that includes 77 kinds of breakfast cereals. The data collected is based on nutritional information and consumer ratings.

The consumer rating is a rating of a cereals "healthiness" for consumer information and is not rated by the consumers. The data set includes 13 numerical variables. As seen in the metadata dictionary below:

	A
1	Source: DATA ANALYSIS FOR STUDENT LEARNING (DASL)
2	
3	1. Name: Name of cereal
4	2. mfr: Manufacturer of cereal where A = American Home Food Products; G = General Mills; K = Kelloggs; N = Nabisco; P = Post; Q = Quaker Oats; R = Ralston Purina
5	3. type: cold or hot
6	4. calories: calories per serving
7	5. protein: grams of protein
8	6. fat: grams of fat
9	7. sodium: milligrams of sodium
10	8. fiber: grams of dietary fiber
11	9. carbo: grams of complex carbohydrates
12	10. sugars: grams of sugars
13	11. potass: milligrams of potassium
14	12. vitamins: vitamins and minerals - 0, 25, or 100, indicating the typical percentage of FDA recommended
15	13. shelf: display shelf (1, 2, or 3, counting from the floor)
16	14. weight: weight in ounces of one serving
17	15. cups: number of cups in one serving
18	16. rating: a rating of the cereals calculated by Consumer Reports
19	
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Figure 1: Metadata dictionary for “Creals.csv”

A dendrogram is a treelike diagram that summarizes the process of clustering. At the bottom are the observations. Similar observations are joined by lines whose vertical length reflects the distance between the observations.

Step One:

Access Google Colab at: https://colab.research.google.com/. Select the blue 'Sign in' button from the top right corner of the page to login.

Note: *You will need a google account to be able to utilize the cloud platform.*

Step Two:

In Google Colab, select *File* from the top navigation bar and then *new notebook* to be able to install the Python Packages required for the lab.

- Rename your notebook as YourLastName_FirstName_LAB# (example: Daniels_Jerome_LAB1).
- Save a copy in your Drive by selecting 'File' and then 'Save a copy in Drive.'

Python Packages

```
# ---import all modules---
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
from scipy.cluster.hierarchy import dendrogram, linkage
import scipy.cluster.hierarchy as sch
#add more python packages, as you need to do the work
```

Step Three:

Use Python in Google Colab to download your dataset "Cereals.csv", to explore and summarize the data as follows:

- Remove all cereals with missing values.
- Run the k-means clustering method. Unlike classification methods, Clustering methods group data instances based on common characteristics.
- Apply hierarchical clustering to the data.
- Compare dendrograms from single linkage and complete linkage and look at the cluster centroids.

- How many clusters would you use?

Rubric

Refer to your Canvas lab assignment for the rubric.

Submission

After you complete each assignment, you will need to share your completed work by uploading an ‘ipynp’ and ‘PDF’ file of your work for your assignment submission. Follow the directions as outlined in the Colab Setup Instructions and follow the sharing section.

1. Name your Colab file: “YourLastName_FirstName_LAB4.ipynp”
2. Show all work as a PDF: “YourLastName_FirstName_LAB4.pdf”

Reference

Plotly. (n.d.). Graphing Libraries. Dendrograms in Python.